Mohammed Pathariya

linkedin.com/in/mjpathariya | mjpathariya.com | github.com/MohammedPathariya
mjpathariya7@gmail.com | +1 (930) 333-7212 | California, USA

AI Engineer building **agentic systems** and **LLM pipelines** for data-driven decision automation | **30x** pipeline speedup, **89%** LLM precision on custom gold set, **80%** agentic benchmark pass rate | **IEEE published**.

Education

Master of Science in Data Science, Indiana University — Bloomington, IN, USA
GPA: 3.75/4.0 | Coursework: LLMs, Deep Learning, Statistical ML, MLOps, Cloud Computing — *Aug. 2024 – May 2026*

BE in Artificial Intelligence & Data Science, Pune University — Pune, MH, India
GPA: 3.80/4.0 | ML, Data Structures & Algorithms, Statistics, Probability & Inference — *Aug. 2020 – May 2024*

Experience

Graduate Research Assistant — Jun. 2025 – May 2026
Indiana University Bloomington — *Bloomington, IN*

- **NBA Analytics Pipeline:** Built an end-to-end pipeline processing **700+ games**, parallelizing **WhisperX** transcription across **5 Slurm-managed HPC nodes**, achieving a **30x speedup** (3.5m → 7s) at near real-time latency.
- Deployed **8-bit quantized Llama-3 8B** (**bitsandbytes**) on **Big Red 200** A100 nodes for psychological marker extraction, achieving **89% precision** vs. a **72% RoBERTa** baseline, validated on a 500-quote gold set.
- Architected a **Transformer Encoder** fusing Llama-3 sentiment embeddings with rolling box-score metrics, achieving **53.8%** Against-The-Spread accuracy on a **120-game** chronological holdout (vs. **52.4%** breakeven).
- **Genomic Edge Pipeline:** Built a custom **alignment-free k-mer classifier** for gene-region assignment (PR/RT/IN) achieving <**1 sec** per-sample on CPU, enabling offline deployment on **NVIDIA Jetson AGX Orin**.

Data Engineering Intern — Jan. 2024 – Jun. 2024
Sparkwood IT Solutions — *Pune, India*

- Collaborated with senior engineer to conduct root-cause analysis of a **5% KPI discrepancy** between Marketing and Finance, standardizing the Net Sales metric via a validated **SQL View** approved for production deployment.
- Optimized legacy **PostgreSQL** reporting queries by implementing composite indexing and refactoring joins, reducing report generation latency by **40%** (5m → 3m).
- Diagnosed recurring **database lock contention** in production **Airflow** ETL pipelines, implementing automated retry logic that reduced nightly pipeline failures by **90%** and ensured reliable daily reporting for business stakeholders.

Projects

The Digital Forge | *Multi-Agent Systems, AI Engineering* | Code — May 2025 – Aug. 2025

- Architected a **sequential multi-agent system** (CrewAI) mimicking the SDLC across 4 specialized agents, achieving a **55% first-shot pass rate** vs. a **45% zero-shot GPT-4** baseline on a **20-task benchmark**.
- Engineered a **self-healing Docker** sandbox capturing **stderr** from failed containers and autonomously re-prompting agents within a 3-strike circuit breaker, salvaging an additional **25%** of tasks for an **80% final pass rate**.
- Implemented a **RAG** layer using **ChromaDB** for semantic API documentation retrieval, grounding agent decisions in verified references and directly rescuing **15%** of tasks from catastrophic library hallucinations.

LearnLoop | *System Design, RAG Engineering* | Code — Jan. 2025 – Apr. 2025

- Designed a **Session-Based RAG** architecture using in-memory **FAISS** with **all-MiniLM-L6-v2** embeddings and **512-token fixed chunking**, achieving **~90% Recall@5** and <**50ms** retrieval latency per user.
- Optimized database concurrency by enabling **Write-Ahead Logging (WAL)**, validated via **Locust** load testing to support **500+** simultaneous users without locking failures.
- Developed a deterministic output parser using **Pydantic**, creating a self-healing feedback loop that autonomously corrects invalid JSON from GPT-4, achieving **99%** UI rendering stability.

AudioGroove | *Deep Learning, Research* | Code — Oct. 2023 – Apr. 2024

- Published **Tunes by Technology** in **IEEE ICC Robins**, statistically validating **Bi-LSTM** outperforms DCGAN by **85%** in structural coherence for symbolic music generation.
- Architected a distributed **Dask** ETL pipeline for **175K+ MIDI files** (20h → 2h) and tracked **50+ experiments** via **MLflow**, identifying optimal hyperparameters to achieve a validation loss of **0.78**.

Technical Skills

Languages & Frameworks: Python, SQL, Bash, Flask, FastAPI, Pydantic, Streamlit, PostgreSQL
AI & GenAI: PyTorch, Hugging Face, LangChain, CrewAI, LlamaIndex, FAISS, ChromaDB, Scikit-learn, OpenAI API
Data & Infrastructure: Docker, GitHub Actions, MLflow, Apache Airflow, Dask, Locust, AWS (EC2, S3), GCP (Vertex AI)
Concepts & Architecture: RAG, Multi-Agent Systems, Agentic Workflows, Explainable AI, Prompt Engineering, CI/CD, TDD
Deployment & Optimization: ONNX, TensorRT, Inference Optimization, bitsandbytes, Slurm, HPC